# Policy Evaluation - Natural Experiments & the Oregon Medicaid Study

April 7, 2026

## Estimating Treatment Effects Review

- $ATE = Avg_n[Y^1 - Y^0]$
- $ATE_{est} = Avg_n[Y^1|D=1] - Avg_n[Y^0|D=0]$
- When $(Y^1, Y^0) \not\perp D$:

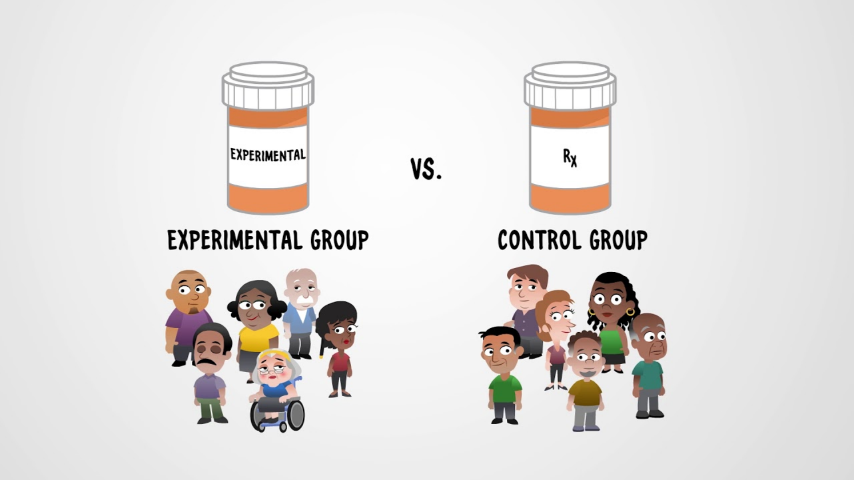
$$ATE_{est} = ATE + \underbrace{\{Avg_n[Y^0|D=1] - Avg_n[Y^0|D=0]\}}_{\text{Selection Bias}}$$
$$+ \quad \underbrace{(1-\pi)(ATT - ATU)}_{\text{Heterogeneous Treatment Effect Bias}}$$

- Randomize?
- $ATE_{est} = \beta_0 + \beta_1 D + \beta_2 X_1 + \beta_3 X_2 + ...\beta_k X_{k-1} + \varepsilon$
- Matched regression

# Natural Experiments to Estimate Treatment Effects

- What is a natural experiment?

# Natural Experiments to Estimate Treatment Effects



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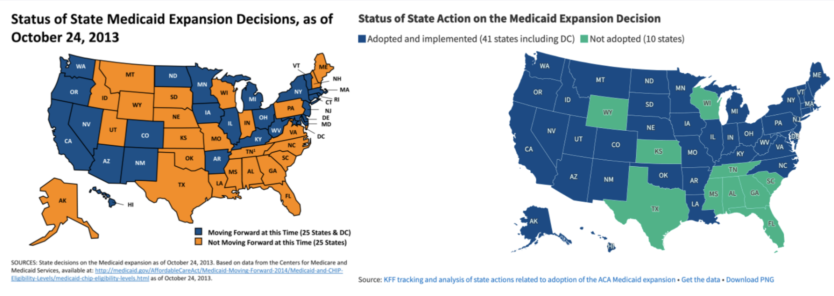

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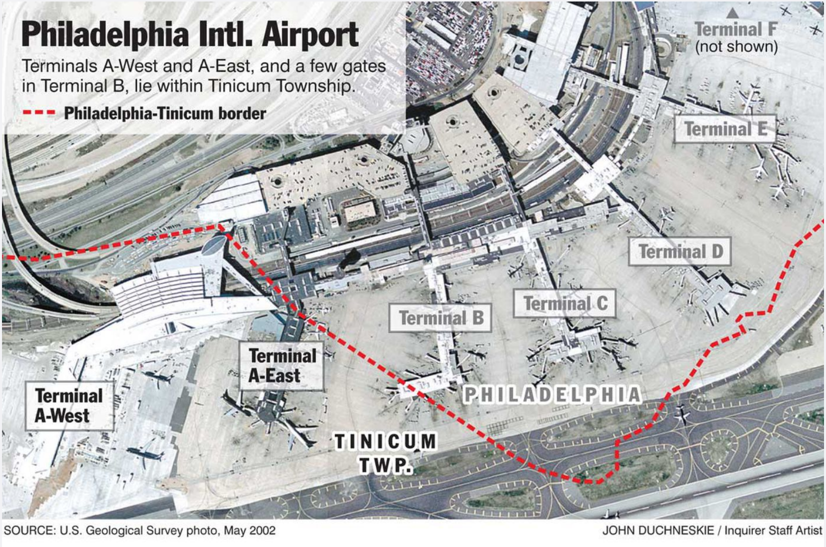
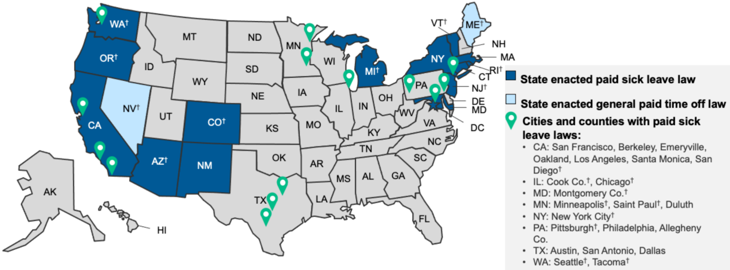


# Natural Experiments to Estimate Treatment Effects

## State and Local Paid Sick Leave Laws, 2021




†Law permits use of accrued leave for workplace closure or closure of the worker's child's school or childcare associated with a public health emergency.

NOTES: NM's law takes effect July 1, 2022. CO's law for employers with fewer than 16 workers takes effect Jan. 1, 2022; currently in effect for all other CO employers. Allegheny Co.'s law was enacted in Sept. 2021 and will take effect 90 days after the county posts compliance information for employers. The three local laws passed in TX are on hold due to a pending court challenge. All other state and local laws are currently in effect. All state and all local paid sick leave laws except Pittsburgh, Oakland, and Berkeley permit use of paid leave for reasons associated with sexual assault, domestic violence, or stalking, known as "safe time."

SOURCE: KFF analysis of state paid family and medical leave laws; A Better Balance. Overview of Paid Sick Time laws in the United States.

KFF

The NEW ENGLAND JOURNAL of MEDICINE

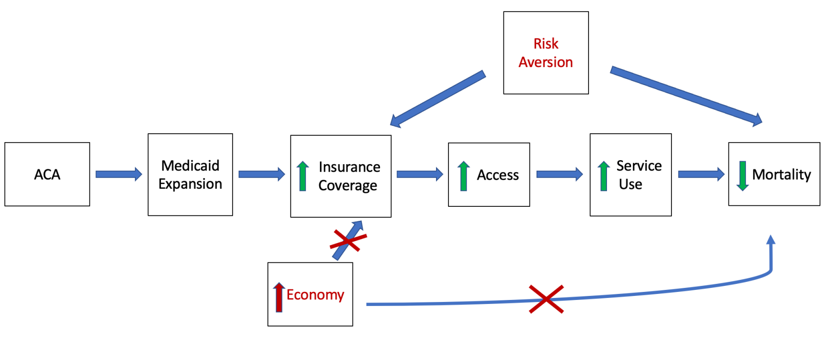
SPECIAL ARTICLE

### Cancer Screening after the Adoption of Paid-Sick-Leave Mandates

Kevin Callison, Ph.D., Michael F. Pesko, Ph.D., Serena Phillips, Dr.P.H., and Julie A. Sosa, M.D.

# Natural Experiments to Estimate Treatment Effects

- Why are natural experiments valuable when estimating treatment effects?



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- OHP Standard enrollment:
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  - 2008: 19k
- Expand OHP Standard enrollment by 10k in 2008.
  - 90k people applied

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OREGON
HEALTH STUDY

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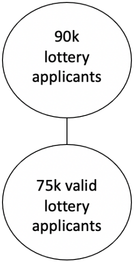

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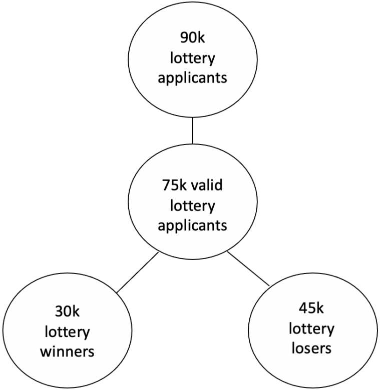

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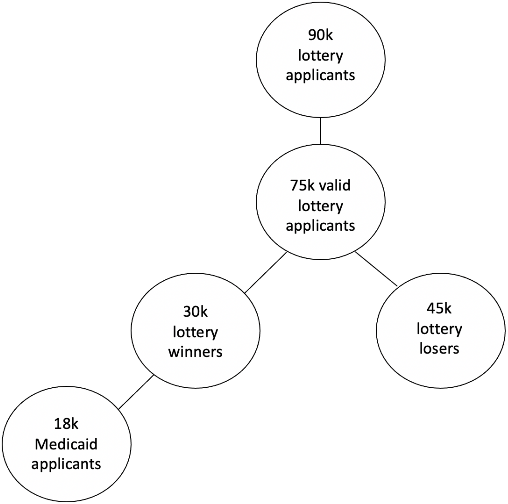

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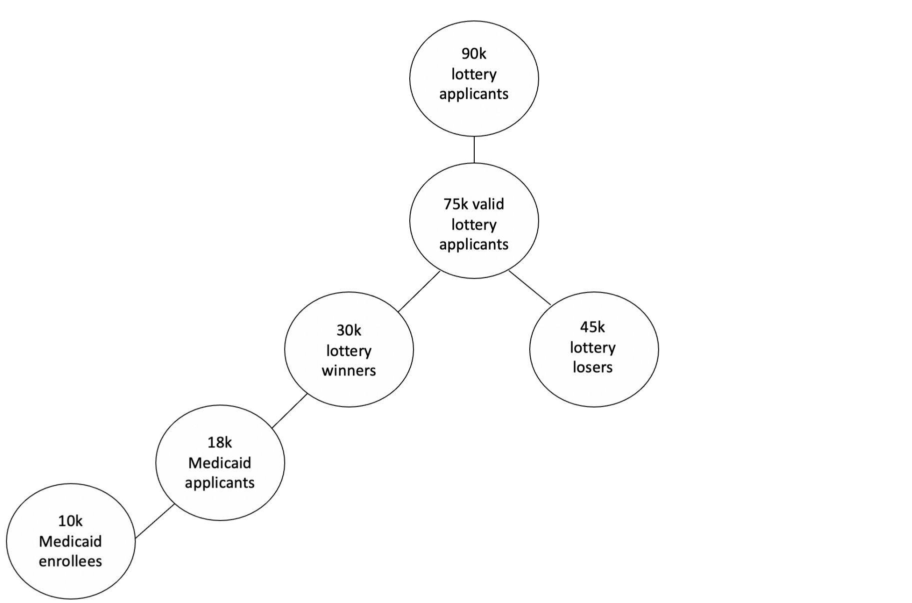

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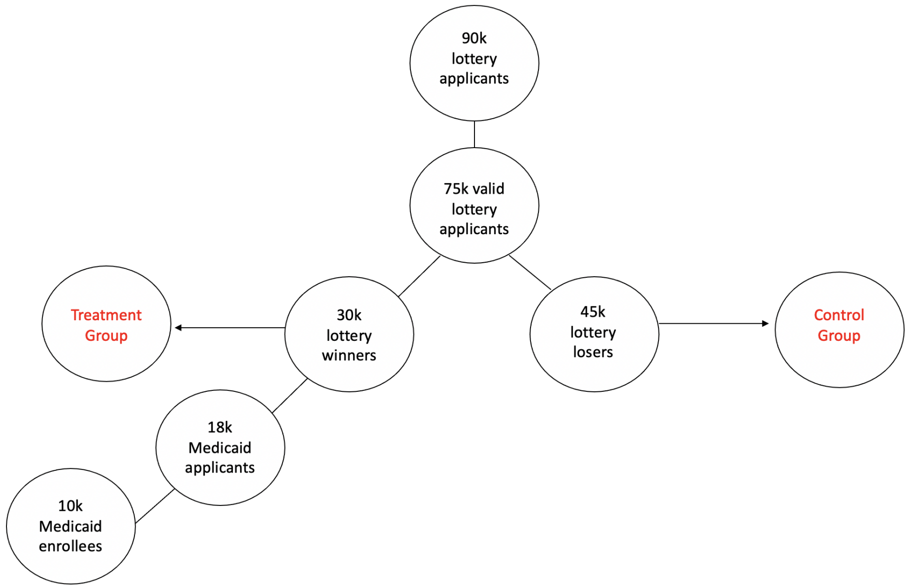

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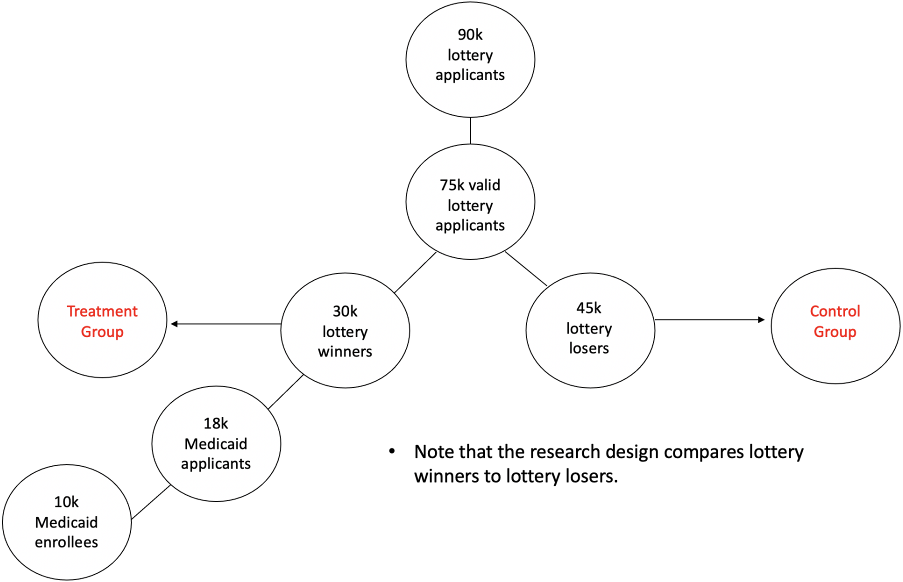


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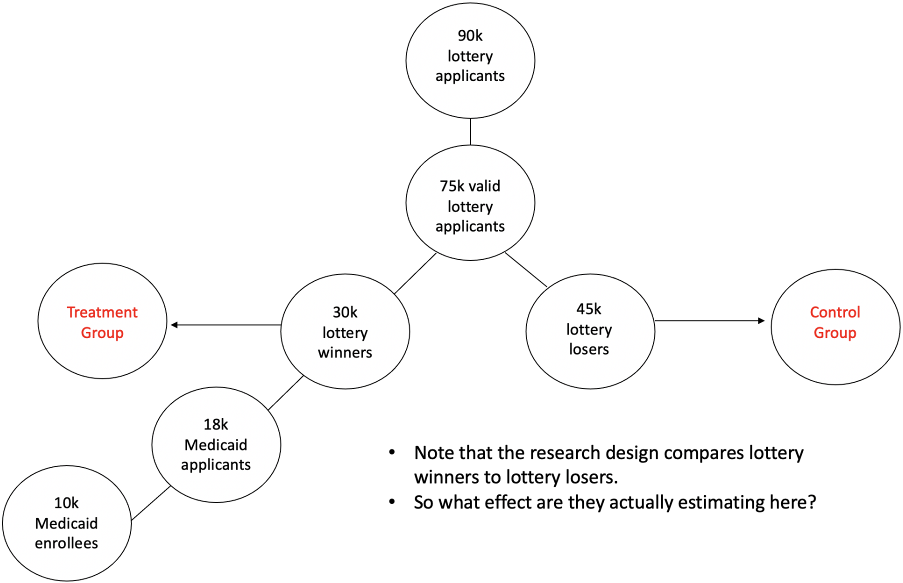


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- So what effect are they actually estimating here?

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  2. Instrumental variables estimate of the local average treatment effect (LATE).

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- **Instrumental Variables - Two-stage least squares**
- First stage: $Insurance_i = \alpha_0 + \alpha_1 Lottery_i + \varepsilon_i$
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  - LATE: those who obtain insurance after winning the lottery and who would not have obtained insurance without winning the lottery.
    - How does this distinction between ITT and TOT/LATE relate to the concept of external validity?